Zhenkui Zhou

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Education

Peking University | BSc in Physics Sep 2024 - Jul 2028 expected

Cumulative GPA: 3.667/4.000

Selected coursework: Probability and Statistics (98/100); Theoretical Mechanics (95); Modern Physics (91); Optics (90); Methods of Mathematical Physics II (90); Linear Algebra (89).

Research Interests

Computational condensed matter physics; molecular and phonon Berry curvature; lattice dynamics; phonon thermal Hall transport.

Research Experience

Berry Curvature and Phonon Thermal Hall Transport Apr 2026 - present

Undergraduate research training | Adviser: Professor Ji Feng, Peking University

- Reviewed the theoretical connection between molecular Berry curvature, phonon dynamics, phonon Berry curvature, and thermal Hall response through literature reading and discussion.
- Focused reading includes Qin, Zhou, and Shi, Phys. Rev. B 86, 104305 (2012), and Saparov et al., Phys. Rev. B 105, 064303 (2022).
- Completed an initial synthesis of the theoretical framework; current work is at the literature and method preparation stage, with numerical implementation as a next goal.

Selected Projects

Climbing Magnet Experiment and Preliminary Modeling

- Worked in a three-person team on a rotating magnetic rod experiment; extracted motion trajectories with Tracker and fitted data using Origin.
- Contributed to a simplified force model and MATLAB visualization, examining limitations associated with friction and asymmetric contact.

QPDS | Monte Carlo Decision Modeling in Poker

- Developed a Python application using Monte Carlo sampling and limited exact enumeration to estimate hand equity and compare expected values, with AI-assisted coding.
- Used pytest for numerical and boundary checks, and Flask to separate the calculation module from the web interface.

Technical Skills

Programming and analysis: Python (NumPy, pandas, Matplotlib, pytest), MATLAB; basic C++.
Tools: Git, Linux, Tracker, Origin; AI-assisted coding workflows.

Honors and Activities

Qin Wanshun and Jin Yunhui Scholarship; Merit Student, Peking University (2025-2026).
Academic Excellence Award, Peking University (2024-2025).
Silver Medal, Chinese Physics Olympiad national final (Oct 2023).
Peking University wushu team; class representative and student administrative assistant.